Melis Kılıç

AI & Data Engineer
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About Me

AI and Data Engineer with production experience building LLM-backend services, agentic workflows, and governed data systems. Designs and ships AI systems end to end: from LangGraph orchestration and prompt engineering to evaluation pipelines, FastAPI backends, and observable infrastructure. Strong focus on structured outputs, tenant-safe data access, and measurable AI quality.

Professional Experience

AI & Data Engineer, QKare Information Technologies, Sivas 2025 to Present

B2B HR analytics SaaS platform combining workforce data, psychometric assessments, and LLM-backed product features. Worked across agentic AI services, governed data layers, evaluation pipelines, and FastAPI backend infrastructure.

- Shipped multi-agent HR analytics and coaching features in production by building LangChain and LangGraph services with FastAPI backends and structured outputs, enabling natural-language workforce insights for HR workflows
- Enabled tenant-safe conversational analytics by designing a governed LangGraph platform with request-scoped PostgreSQL RLS-aware tool calls, separated domain contracts across analysis modules, and YAML-backed prompt configuration versioned in Git
- Made AI Coach quality measurable by building an LLM evaluation workflow: 49 POML orchestrator prompts across 43 coaching modes, simulated LLM-to-LLM sessions, and LLM-as-judge scoring across 5 quality dimensions with OpenAI Evals-compatible JSONL exports
- Made KPI recommendations auditable before business adoption by grounding a position-based recommendation service in 2,680 APQC metrics and 1,631 PCF nodes; and adding LLM-as-judge SMART-alignment scoring
- Powered tenant-safe Apache Superset BI dashboards by modeling 56 mart tables through a 109-model dbt Core layer, resolving critical employee-level aggregation errors
- Ensured end-to-end data reliability by enforcing PostgreSQL RLS tenant boundaries as testable pipeline contracts and adding automated parity tests to catch data bugs.
- Reduced per-agent setup overhead across an AI service monorepo by building a reusable FastAPI gateway factory with standardized auth, structured JSON logging, correlation IDs, OpenTelemetry hooks, and cookiecutter scaffolding with multi-stage Docker builds

AI Research Intern, Inncrea Software & IT, Sivas Aug 2024 to Sept 2024

- Built repeatable CT image classification experiments with fold-based evaluation, augmentation pipelines, and CNN backbone benchmarking for stroke detection workflows
- Reduced inference cost while preserving diagnostic performance by distilling InceptionV3 teacher knowledge into lightweight EfficientNet-B0 student models, forming the foundation of the open-source Stroke Classification project

Education

Sivas Cumhuriyet University, Bachelor's Degree, Computer Engineering, Turkey 2021 to 2025
AGH University of Krakow, Erasmus+ Exchange, Computer Engineering, Poland 2023 to 2024
National Technology Academy, AI Specialization Program Dec 2024 to Jun 2025

Skills

Backend & API: Python, FastAPI, Pydantic, REST APIs, WebSocket, Redis
Data Engineering: dbt Core, PostgreSQL, Row-Level Security (RLS), Apache Superset, SQL, ETL/ELT
AI & LLM Systems: LangChain, LangGraph, Langfuse, OpenAI API, RAG, Agentic AI, Prompt Engineering
ML & Evaluation: PyTorch, scikit-learn, Hugging Face, ONNX, LLM-as-Judge, OpenAI Evals
Infra & DevOps: Docker, Argo Workflows, GitLab CI/CD, OpenTelemetry, pytest

Projects

Stroke Classification, Accepted by TEKNOFEST Healthcare AI 2024 to 2025

- Outperformed a heavier InceptionV3 teacher by distilling its knowledge into a compact EfficientNet-B0 student; soft-voting ensemble reached 98.2% accuracy/F1 on 49,500 CT images across 3 cross-validation folds
- Built a YAML-configured MLOps pipeline with uv for preprocessing, augmentation, class balancing, and stratified 3-fold validation; published model weights, a live Streamlit demo on Hugging Face

Additional Information

Languages: Turkish (Native), English (Professional Working Proficiency)
Certificates: HCCDA-AI (Huawei Certified Developer Associate – AI), Huawei;
Artificial Intelligence Basic Training Certificate, National Technology Academy